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A tractable two-stage robust winner determination model for truckload service procurement via combinatorial auctions

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ABSTRACT

A combinatorial auction is one of the adopted mechanisms for truckload (TL) service procurement. In such an auction, the shipper faces a well-known winner determination problem (WDP): the shipper, as the auctioneer, is given bids submitted by a group of carriers. In most literature, WDP is modeled as a deterministic mixed-integer program (MIP) and is solved by standard MIP algorithms. However, in practice, the exact shipping demand is unavailable until after the auction. This shipment volume uncertainty has a significant impact on the solution to WDP. Therefore, a deterministic winner determination model with an estimate of shipment volume may not provide solutions that attain low procurement costs. This paper proposes a new tractable two-stage robust optimization (RO) approach to solve WDP for TL service procurement under shipment volume uncertainty. Assuming that only historical data is available, we propose a data-driven approach based on the central limit theorem (CLT) to construct polyhedral uncertainty sets. In particular, we consider two random cases: independent shipment volume and correlated shipment volume. A two-stage RO model with integer first-stage decision variables and continuous recourse variables is then formulated. We develop a reformulation solution method and use numerical tests to demonstrate that it is much more computationally efficient than the widely adopted Benders' type constraint generation algorithm. We demonstrate by numerical tests that real-world sized instances of TL service procurement problems can be solved by our proposed robust method. Moreover, we compare our robust approach with benchmark and show that it is more tractable and robust to uncertainty.

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1. Introduction

Truckload (TL) transportation is the movement of homogenous freight for a single shipper between a fixed origin and a fixed destination. It is a service provided by TL carriers. The shipper, who is the owner of the freight, can be an agent at any stage of a supply chain (e.g., suppliers, manufacturers, distributors, or retailers). TL transportation is a necessary component of a shipper's logistics system and the associated expenditure accounts for a significant portion of the shipper's overall cost. In many corporations, transportation expenditure accounts for about 30% of the overall cost of goods sold (Chen et al., 2009; Ballou, 1992). From a market perspective, truck transportation accounts for 29% of for-hire transportation services

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expenditure which is about 363.7 billion U.S. dollars in 2007 (RITA, 2009). TL accounts for over half of total trucking expenditure (Sheffi, 2004). Reducing TL service procurement cost can greatly benefit large shippers.

By late 1990s, most shippers were buying transportation service by a request-for-proposal (RFP) process via an auction (Sheffi, 2004). The process documented by Foster and Strasser (1990) is the following: the shipper first estimates the shipping demand on each origin–destination (OD) pair in the coming year. Then it transforms this information into a list of lanes[1] and passes it to every carrier that is invited to the auction. Each carrier quote prices on lanes, and the shipper finally evaluate the bids and selects the winners of each lane. Even nowadays in China's fast-growing logistics market, many fast-moving consumer goods and bulk stock manufacturers still procure transportation services through such a RFP process.[2] However, this standard process inherently neglects lane interdependencies which usually lead to each carrier quoting a higher price on each lane, it has been gradually replaced by combinatorial auction which allows the carriers bidding on packages of lanes[3] (Sheffi, 2004).

Determining the winners of a combinatorial auction so that the total procurement cost to the shipper is minimized is nontrivial. Optimization-based methods have been proposed to tackle the problem (see Caplice and Sheffi (2003) for an example). Since exact volume of shipments in the coming year is not available at the time that the combinatorial auction is conducted for TL service procurement, usually the shipper forecasts the volume of shipments on each lane and conducts the auction with that forecast. However, by this method, when the shipper determines the winners, they do not take into account the costs due to the errors in the predicted shipment volume. For instance, due to limited carrier capacities, if the realized shipment volume is larger than forecasted, it is possible that additional TL service is required from third-party carriers who did not participate in the auction. Thus, extra costs which are not accounted for in the deterministic WDP will be incurred. Therefore, robust winner determination methods are needed that can characterize and mitigate the impact of shipment volume uncertainty on TL service procurement.

Recently, a new bidding structure and a corresponding two-stage stochastic winner determination model for TL service procurement was proposed by Ma et al. (2010). Assuming that the distribution of uncertain shipment volume is exactly known to the shipper, Ma et al. (2010) is a remarkable first attempt to explicitly model shipment volume uncertainty and resolve the corresponding stochastic WDP. However, in many cases the exact distribution of uncertainty may not be available. In fact, often decision maker cannot fit the historical data into any common probability distribution but derive some statistical estimates such as mean and variance. So a method that requires no more than this partial distributional information of uncertainty will be more practical. What is more, as reported in Zhang et al. (2014), sampling-based stochastic WDP can become computationally inefficient when the problem scale is large. So it is desirable to use a numerically tractable method to solve WDP under uncertainty.

Robust optimization (RO) derives optimal solutions in worst-case scenario under uncertainty (Soyster, 1973; El Ghaoui et al., 1997, 2003; Ben-Tal and Nemirovski, 1998, 1999, 2000; Bertsimas and Sim, 2003, 2004). It serves as an alternative to the classical stochastic programming methods that handles mathematical programs under uncertainty. Recently, Bandi and Bertsimas (2012) propose to apply RO to several stochastic systems including queuing networks, auction design, and option pricing. The authors demonstrate their approach is numerically tractable, while the classical approach to these application areas that relies on probability theory-based analysis is often intractable. Inspired by this idea, in order to retain tractability while handling uncertainty, in this paper, utilizing the first and second moments of uncertainty which are always estimable from historical data, we propose a tractable two-stage robust winner determination model.

The main contributions of this paper are summarized as follows:

- This paper refines the bidding structure proposed by Ma et al. (2010) to better characterize the effects on the solution to WDP brought by shipment volume uncertainty. In particular, shortage in carriers' required shipment volume is allowed but will be penalized. Moreover, the model allows multiple carriers to win the same single lane which provides a shipper a higher degree of feasibility to select the winners.
- This paper proposes a data-driven approach to develop a two-stage robust winner determination model. Specifically, unlike most two-stage RO studies, we apply a central limit theorem (CLT) based approach to construct uncertainty sets. By this approach, only mean and variance which are estimable from historical data are needed and we can handle either independent or correlated random shipping demand. We develop a reformulation solution method that is much more efficient than the constraint generation algorithm which is widely adopted in two-stage RO literature.
- This paper applies the proposed method to real-world sized instances of TL service procurement. Numerical examples are provided to demonstrate the value of the proposed two-stage robust winner determination model. Specifically, we use numerical experiments to show that the proposed robust WDP is even more numerically tractable than the deterministic WDP. Moreover, by simulation, we show the RO solution is more robust to shipment volume uncertainty and can result in much lower procurement cost than the benchmark especially when the variance of uncertainty is large.
- The two-stage robust winner determination model proposed in this paper has more potential impacts than the benchmark on TL procurement in practice. The intuition for this insight is threefold. First, in many application scenarios, mean and variance are the only solid statistics available from historical data. The proposed CLT based approach just requires

[1] We use the term "lane" to refer to an "O–D-commodity-period specific flow" as first described by Jara-Diaz (1988).

[2] This is concluded from the first author's experience in interacting with both shippers and carriers in China's logistics market.

[3] In a package bid, the carrier quotes one price on multiple lanes. Usually the lanes in a package will help a carrier form circle trips which can increase backhaul efficiency and cut the operating cost.

mean and variance to construct uncertainty set. Second, in real-world TL service auctions, lane demands are usually highly correlated. The proposed model incorporates demand correlation and generates solutions which are more robust to uncertainty than benchmark. Third, the proposed robust model is computationally efficient.

The rest of the paper is organized as follows. In Section 2, we provide a literature review. In Section 3, we present a refined deterministic formulation of the WDP for TL service procurement. The tractable two-stage robust WDP is then proposed in Section 4. A reformulation solution method is developed in Section 5. We conduct numerical experiments to analyze the results in Section 6. In Section 7, we relax one modeling assumption and discuss its effects on our study. In the end, Section 8 concludes the paper and discusses the future work.

2. Literature review

A combinatorial auction is a mechanism in which bidders can bid on combinations of items. It is often conducted in the sale of assets such as Federal Communications Commission (FCC) spectrum, airport time slots, pollution permits, railroad segments, and network routing (see de Vries and Vohra (2003) for a general survey and Kuo and Miller-Hooks (2012) for an example in railroad segments). In transportation service procurement, many large shippers (e.g., Wal-Mart Stores Inc., The Home Depot Inc., K-Mart Corporation) have been utilizing combinatorial auctions to procure TL services from carriers (see Sheffi (2004) for a list of shippers that procure TL service via combinatorial auctions). From the carriers' perspective, the creation of package bids and strategic bidding are key research topics (e.g., Song and Regan (2005), Lee et al. (2007), Chang (2009) and Triki et al. (2014)). From the shippers' perspective, the winner determination problem (WDP) is crucial. Recently, from a neutral perspective, Huang and Xu (2013) and Xu and Huang (2014) investigate the auction design for transportation service procurement to increase both shippers' and carriers' utilities.

WDP refers to the process of allocating items to bidders in a combinatorial auction that optimizes the auctioneer's utility given a set of bids, which is an NP-hard integer program (Lehmann et al., 2006). A variety of mathematical formulations and numerical solutions of WDP have been developed (Rothkopf et al., 1998; Sandholm, 1999; Andersson et al., 2000; Nisan, 2000; de Vries and Vohra, 2003). A detailed review of WDP can be found in Lehmann et al. (2006). In the context of transportation procurement, Caplice and Sheffi (2003) develop several integer programming formulations of WDP with side constraints that reflect business considerations from both shippers and carriers.

There are a few studies which address TL procurement under uncertainty. Assuming that repositioning capacities and costs are uncertain, Chen (2010) studies how carriers should generate bids to maximize their utility and how the resulting combinatory auction could be solved. Ma et al. (2010) propose the first model that addresses WDP under shipment volume uncertainty. In particular, a new bidding structure which takes into account the uncertain shipment volume on each O–D pair is proposed and a two-stage stochastic program with recourse is developed to minimize the expected procurement cost. The first-stage decision variable is binary and represents the allocation of lanes to the carriers, while the second-stage recourse variables are the assignments of freight to the winning carriers and thus are continuous. In Ma et al. (2010), it is assumed that the uncertain shipments will realize at 1 of the 3 levels (low, medium and high) with specified probability and volume. Therefore, the two-stage stochastic program is actually equivalent to a deterministic mixed integer program (MIP) and can be solved by commercial software packages like CPLEX. As an extension of Ma et al. (2010), Zhang et al. (2014) develop a sampling-based two-stage stochastic programming approach which can deal with shipment volume uncertainty that has a continuous probability distribution.

Robust optimization (RO) has been developed in the past 15 years (Ben-Tal and Nemirovski, 2002, 2008, 2009; Bertsimas and Sim, 2003, 2004; Ben-Tal et al., 2004). For a comprehensive review of methodology and applications of robust optimization, we refer the readers to Ben-Tal et al. (2009) and Bertsimas et al. (2011). Among the RO methods, multi-stage RO, especially two-stage RO, has been successfully applied to a variety of problems including network design, inventory routing, and the design and operations of power grid systems. Specifically, Ben-Tal et al. (2011) propose to apply an affinely adjustable robust counterpart approach for dynamic emergency logistics planning under demand uncertainty. Thiele et al. (2009) propose a modeling and solution approach to two-stage robust linear program with recourse. Atamtűrk and Zhang (2007) develop a two-stage robust network design model under travel demand uncertainty. Distinct from Thiele et al. (2009), the first-stage decision variable in Atamtűrk and Zhang (2007) is required to be integer. Jiang et al. (2012) propose a two-stage robust unit commitment model for wind power system under wind output uncertainty. A similar study has been also conducted by Bertsimas et al. (2013). In the context of TL service procurement, Remli and Rekik (2013) propose a two-stage robust winner determination model based on the bidding structure and deterministic modeling framework originated by Ma et al. (2010). Most of these two-stage RO studies adopt a constraint generation algorithm that is similar to Benders' decomposition to solve the problem.

Our research was conducted independently of the work in Remli and Rekik (2013). This paper is distinct from Remli and Rekik (2013) in the following perspectives. First of all, our work is based on a refinement of the deterministic model adopted by Remli and Rekik (2013). Our modified deterministic model is more general. Moreover, we assume that only mean and variance of the uncertain demands are available from historical data and construct uncertainty sets by a CLT based approach. By this approach, our model is capable of handling correlated lane demands which as far as we know are ignored in most studies of stochastic or robust WDP for TL service procurement including Remli and Rekik (2013). We develop a new

reformulation solution method which, demonstrated by numerical tests, is much more efficient than the constraint generation algorithm that is adopted in Remli and Rekik (2013) and many other two-stage RO literature. What is more, we evaluate the quality of our solution in a sample of general scenarios by simulation and compare our robust solution with the benchmark. These distinct investigations provide more insights on the tractability and robustness of the solution to WDP under uncertainty and the impact of robust solution on costs of TL service procurement.

3. Deterministic WDP

In this section, we will present a deterministic winner determination model which is based on the following structure of the combinatorial auction for TL service procurement: (1) carriers will submit XOR bids which basically requires that each bidder although can submit multiple bids will win at most one of them (Nisan, 2006); (2) when a carrier submits a package bid, in addition to the price, they can also specify the range of shipment volume that they are willing to accept; (3) based on the bids they collects, the shipper needs to not only determine the winners of the auction but also assign shipments to each winner according to the range specified by the winner; (4) the shipper can also assign shipments to third-party carriers who do not participate in the auction (and usually asks for a higher price); (5) if the volume of shipments assigned to a winner does not meet their specified requirement, a penalty cost on the shipper will be incurred. In addition, for simplifying the analysis, we also assume that a carrier serves each lane in the awarded package to the same level. We will relax this assumption in Section 7 and demonstrate it does not affect our proposed approach.

The notation is summarized in Table 1.

The deterministic WDP (dWDP) can be formulated as:

$$(d\text{WDP})\quad \min \quad \sum_{j\in J}\sum_{k\in K_j}(c_{jk}y_{jk}+p_{jk}z_{jk})+\sum_{i\in I}e_i\varphi_i \tag{1}$$

$$\text{s.t.} \quad \sum_{j\in J}\sum_{k\in K_j}a_{ijk}y_{jk}+\varphi_i=d_i,\quad i\in I \tag{2}$$

$$LS_{jk}x_{jk}\leqslant y_{jk}+z_{jk},\quad j\in J,\ k\in K_j \tag{3}$$

$$US_{jk}x_{jk}\geqslant y_{jk},\quad j\in J,\ k\in K_j \tag{4}$$

$$\sum_{k\in K_j}x_{jk}\leqslant 1,\quad j\in J \tag{5}$$

$$\sum_{j\in J}\sum_{k\in K_j}x_{jk}\leqslant N_{\max} \tag{6}$$

$$\sum_{j\in J}\sum_{k\in K_j}x_{jk}\geqslant N_{\min} \tag{7}$$

$$x_{jk}=\{0,1\},\qquad j\in J,\quad k\in K_j \tag{8}$$

$$y_{jk},\quad z_{jk},\quad \varphi_i\geqslant 0\quad i\in I,\quad j\in J,\quad k\in K_j \tag{9}$$

In (dWDP), Eq. (1) is the objective function which minimizes the total procurement cost to the shipper. Specifically, the total cost consists of the service cost, extra cost for using third-party carriers, and a penalty cost for failing to satisfy the winning carriers' requirement. Eq. (2) is a constraint that guarantees the shipping demand on each lane is satisfied either by the carriers that participate in the auction or third-party carriers who are not invited to the auction but are still able to provide

Table 1
Notations.

Symbol	Description
I	The set of lanes
J	The set of carriers
K_j	The set of packages submitted by carrier j
d_i	The volume of shipments on lane i
a_{ijk}	An indicator parameter that equals 1 if lane i is included in carrier j's package k and 0 otherwise
$N_{\min}$	Minimum number of winners of the auction specified by the shipper
$N_{\max}$	Maximum number of winners of the auction specified by the shipper, $N_{\max}\geqslant N_{\min}$
LS_{jk}	The minimum volume of shipments on all lanes in package k that is required by carrier j if carrier j wins package k
US_{jk}	The maximum volume of shipments on all lanes in package k that is required by carrier j if carrier j wins package $k, US_{jk}\geqslant LS_{jk}$
e_i	Extra service cost for shipping 1 unit of freight by third-party carriers who are not invited to the auction
p_{jk}	The penalty cost of being in short of 1 unit of shipment on every lane in winner j's package k
c_{jk}	The price of shipping 1 unit of freight on all lanes quoted by carrier j on their package k
x_{jk}	A binary decision variable and it equals 1 if carrier j wins her package k and 0 otherwise
y_{jk}	Volume of shipments assigned to carrier j on all lanes in their package k
z_{jk}	The shortage in required shipment volume on all lanes in carrier j's package k
φ_i	Volume of shipments on lane i that is assigned to third-party carriers

truckload service. The minimum and maximum volumes of required shipments on all lanes in each package of each winning carrier are characterized by Eqs. (3) and (4), respectively. Eq. (5) represents the XOR bidding language. Eqs. (6) and (7) reflect the restrictions on the number of winning carriers which are often adopted by shippers to determine the optimal size of the core carrier group (Caplice and Sheffi, 2003). Eqs. (8) and (9) are integrality constraint and nonnegativity constraint, respectively.

Compared to the deterministic model proposed in Ma et al. (2010), (*d*WDP) is more general and robust to shipment volume uncertainty. Specifically, with the shortage in required shipment volume explicitly modeled in constraint (3), (*d*WDP) is more general. Moreover, when the realized shipping demand on a lane is less than the minimum requirement of the carrier that wins the lane, constraint (2) may become infeasible if shortage is not allowed. This issue is overlooked in Ma et al. (2010) and Remli and Rekik (2013). What is more, this model allows multiple carriers winning the same single lane. This will greatly enlarge the feasible region of the shipper's decision pool which leads to an optimal solution that guarantees a lower procurement cost.

4. Two-stage robust WDP

As we mentioned earlier, the volume of shipments on each lane is usually unknown to the shipper at the time of auction. So in (*d*WDP) the parameter d_i in Eq. (2) is a deterministic estimate of the shipment volume on lane *i*. However the error of the estimate and its associated costs are not accounted for in (*d*WDP). In this section, we propose a tractable two-stage robust winner determination model which derives a solution that is robust to uncertainty. Specifically, utilizing mean and variance of the uncertain shipment volume, we construct a polyhedral uncertainty set for shipment volume in Section 4.1 and formulate a two-stage robust winner determination model in Section 4.2. In Section 4.3, we relax the assumption of independent lane demands and elaborate the robust model under correlated lane demand uncertainty.

4.1. Construction of uncertainty sets

We denote the uncertain volume of shipments on lane *i* by d_i and assume that $\tilde{d}_i,\ i \in I$ are random variables. Different from Ma et al. (2010) and Zhang et al. (2014), in this paper we assume that only limited information regarding the distribution of $\tilde{d}_i$, namely mean μ and variance σ^2, is known. For simplicity, we first consider the case when $\tilde{d}_i,\ i \in I$ are independent and identically distributed.[4] We will discuss the correlated lane demands in Section 4.3.

With respect to the classical central limit theorem (CLT), the random variable $\sum_{i\in I}\tilde{d}_i$ follows a standard normal distribution as $\|I\| \to +\infty$. [5] That is,

$$\lim_{\|I\|\to+\infty} Pr\left(\frac{\sum_{i\in I}\tilde{d}_i - \|I\|\mu}{\sigma\sqrt{\|I\|}} \leqslant t\right) = Pr(Z \leqslant t)$$

where $Z \sim N(0,1)$.

Instead of describing $\tilde{d}_i,\ i \in I$ as random variables, we can assume the possible realizations denoted by $\bar{d}_i,\ i \in I$ fall in the following uncertainty set:

$$\Delta = \left\{ (\bar{d}_1, \ldots, \bar{d}_{\|I\|}) \left| \left| \sum_{i\in I}\bar{d}_i - \|I\|\mu \right| \leqslant \Gamma\sigma\sqrt{\|I\|} \right.\right\}$$

where Γ is a constant that usually takes the value of 2 or 3 to make a good fit empirically (Bandi and Bertsimas, 2012).[6]

Note that Δ is rather a different representation than an approximation of the true distribution of $\tilde{d}_i,\ i \in I$, since Δ does not involve any probabilistic factor. However, we still care the quality of Δ which can be reflected by how well the range of $\sum_{i\in I}\bar{d}_i$ matches the tolerance interval of $\sum_{i\in I}\tilde{d}_i$. In a special case when $\tilde{d}_i,\ i \in I$ follows $N(\mu,\sigma^2)$, no matter the size of *I*, the range of $\sum_{i\in I}\bar{d}_i$ is the exact tolerance interval of $\sum_{i\in I}\tilde{d}_i$ (95.45% when $\Gamma = 2$ and 99.73% when $\Gamma = 3$) because $\sum_{i\in I}\tilde{d}_i$ follows normal distribution with mean $\|I\|\mu$ and variance $\|I\|\sigma^2$. If $\tilde{d}_i,\ i \in I$ follows a non-normal distribution, per CLT, as $\|I\|$ approaches infinity, the range of $\sum_{i\in I}\bar{d}_i$ tends to become the exact tolerance interval of $\sum_{i\in I}\tilde{d}_i$. When $\|I\|$ is relatively small, we investigate how the quality of Δ changes with the size of *I* by Monte Carlo simulation. Specifically, we consider two distributions for $\tilde{d}_i,\ i \in I$: unif(50, 150) and Gamma(12, 8.33). Both distributions have same mean and variance which are 100 and 833.33, respectively. As a result, the lower and upper bounds of $\sum_{i\in I}\bar{d}_i$ are $100\|I\| - \Gamma\sqrt{833.33\|I\|}$ and $100\|I\| + \Gamma\sqrt{833.33\|I\|}$,

[4] This assumption can be relaxed as "independent but not necessarily identically distributed" and we can still construct a polyhedral uncertainty set using CLT (see APPENDIX A). In this paper, we keep the original assumption for simplicity.

[5] We use "$\|\cdot t\|$" to denote the cardinality of a set.

[6] Note that Γ in this paper is different from the parameter that controls the conservatism of a budget uncertainty set. In a budget uncertainty set, Γ denotes the maximum number of uncertain coefficients that are allowed to change and can take the value from 0 to the total number of uncertain coefficients (Bertsimas and Sim, 2004).

respectively. We denote the lower bounds by LL and the upper bounds by UU. By setting $\Gamma = 3$, we can calculate LL and UU given the size of I. For each I, we create 10,000 samples of $\tilde{d}_i,\ i \in I$ and denote each sample by $\hat{d}_i$. Then $\sum_{i\in I}\hat{d}_i$ is calculated for each sample and the lower and upper bounds of the 99.73% tolerance interval of the 10,000 samples are recorded. Let's denote the lower and upper bounds of the tolerance interval by $\overline{LL}$ and $\overline{UU}$, respectively. The quality of is measured by

$$R = \frac{|LL - \overline{LL}| + |UU - \overline{UU}|}{\|I\|\mu}.$$

The simulation is replicated for 30 times and how the averaged R changes with the size of I is illustrated in Fig. 1. Clearly, when $\tilde{d}_i,\ i \in I$ follows either distribution, the quality of Δ increases with the size of I.

Since $\bar{d}_i$ stands for shipment volume, it is required to be nonnegative. As a result, we have the following polyhedral set for the uncertain shipment volume:

$$\overline{\Delta} = \left\{ (\bar{d}_1, \ldots, \bar{d}_{\|I\|}) \middle| \left| \sum_{i\in I} \tilde{d}_i - \|I\|\mu \right| \leqslant \Gamma\sigma\sqrt{\|I\|},\ \bar{d}_i \geqslant 0,\ \forall i \in I \right\}.$$

4.2. Two-stage robust formulation of WDP

With the shipment volume described by the uncertainty set $\overline{\Delta}$ specified in the previous section, we can formulate the following two-stage robust winner determination model:

$$(r\text{WDP}) \quad \min_{\mathbf{x}\in X} \max_{\bar{\mathbf{d}}\in\overline{\Delta}} \min_{(\mathbf{y},\mathbf{z},\boldsymbol{\varphi})\in Y(\mathbf{x},\bar{\mathbf{d}})} f_0(\mathbf{y},\mathbf{z},\boldsymbol{\varphi}) \tag{10}$$

where

$$f_0(\mathbf{y},\mathbf{z},\boldsymbol{\varphi}) = \sum_{j\in J}\sum_{k\in K_j}(c_{jk}y_{jk} + p_{jk}z_{jk}) + \sum_{i\in I} e_i\varphi_i,$$

$$X = \{(x_{jk}, j \in J, k \in K_j) |\ (5)\text{–}(8)\}$$

$$Y(\mathbf{x},\bar{\mathbf{d}}) = \{(y_{jk}, z_{jk}, \varphi_i, i \in I, j \in J, k \in K_j) |\ (2)\text{–}(4) \text{ and } (9)\}.$$

In (rWDP), $\mathbf{x}$ and $(\mathbf{y},\mathbf{z},\boldsymbol{\varphi})$ are respectively the vectors of first-stage and second-stage decision variables. In practice, at the time of the auction, the shipper needs to allocate lanes to carriers ($\mathbf{x}$) when the shipping demand on each lane is still uncertain. Once the shipment volume is realized, the shipper needs to determine the volume of shipments that are assigned to each winning carrier ($\mathbf{y}$) and third-party carriers ($\boldsymbol{\varphi}$). In addition, the shortage ($\mathbf{z}$) in shipments that is required by winning carriers is also revealed. By solving (rWDP), we are actually seeking the most robust allocation of lanes to carriers such that the total procurement cost in the worst-case scenario, or in other words the maximum total procurement cost that might be incurred, is minimized.

Note that (rWDP) is a min–max–min problem which is difficult to handle. A model simplification is necessary. Because the inner minization problem is a linear program once the winners are determined and the volume of shipments on each lane is known, we can formulate the dual of the inner minimization problem $\min_{(\mathbf{y},\mathbf{z},\boldsymbol{\varphi})\in Y(\mathbf{x},\bar{\mathbf{d}})} f_0(\mathbf{y},\mathbf{z},\boldsymbol{\varphi})$ in (rWDP), the optimal objective value of which is equal to that of the inner minimization problem due to strong duality theory:

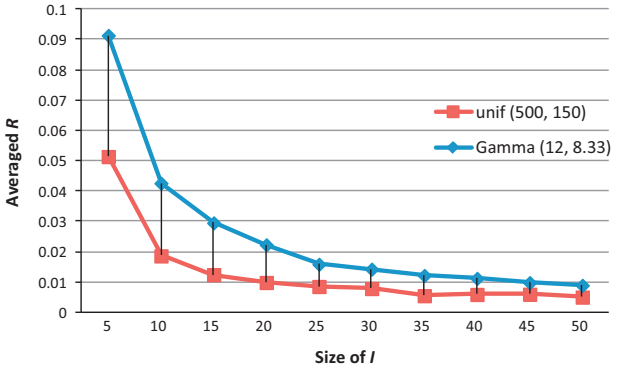


Fig. 1. Averaged R at different sizes of I.

$$(\text{P1})\ \max \quad \left[\sum_{i\in I}\bar{d}_i\lambda_{1,i}+\sum_{j\in J}\sum_{k\in K_j}(LS_{jk}x_{jk}\lambda_{2,jk}-US_{jk}x_{jk}\lambda_{3,jk})\right] \tag{11}$$

$$\text{s.t.}\quad \sum_{i\in I}a_{ijk}\lambda_{1,i}+\lambda_{2,jk}-\lambda_{3,jk}\leqslant c_{jk},\qquad j\in J,\quad k\in K_j \tag{12}$$

$$\lambda_{1,i}\leqslant e_i,\quad i\in I \tag{13}$$

$$\lambda_{2,jk}\leqslant p_{jk},\qquad j\in J,\quad k\in K_j \tag{14}$$

$$\lambda_{2,jk},\ \lambda_{3,jk}\geqslant 0\qquad j\in J,\quad k\in K_j \tag{15}$$

where $\lambda_{1,i},\lambda_{2,jk},\lambda_{3,jk}$ are the dual variables corresponding to constraints (2)–(4), respectively. Then, we can reformulate the max–min problem $\max_{\bar{\mathbf{d}}\in\bar{\Delta}}\min_{(\mathbf{y},\mathbf{z},\boldsymbol{\varphi})\in Y(\mathbf{x},\bar{\mathbf{d}})}f_0(\mathbf{y},\mathbf{z},\boldsymbol{\varphi})$ as:

$$(\text{P2})\quad \max \quad \left[\sum_{i\in I}\bar{d}_i\lambda_{1,i}+\sum_{j\in J}\sum_{k\in K_j}(LS_{jk}x_{jk}\lambda_{2,jk}-US_{jk}x_{jk}\lambda_{3,jk})\right] \tag{11}$$

$$\text{s.t.}\quad \sum_{i\in I}\bar{d}_i\leqslant \|I\|\mu+\Gamma\sigma\sqrt{\|I\|} \tag{16}$$

$$-\sum_{i\in I}\bar{d}_i\leqslant -\|I\|\mu+\Gamma\sigma\sqrt{\|I\|} \tag{17}$$

$$\bar{d}_i\geqslant 0\quad i\in I \tag{18}$$

constraints (12)—(15)

Note that although (P1) and (P2) have the same objective function, the shipping demand $\bar{d}_i$ becomes a decision variable in (P2). If we represent (P2) by $f_1(\mathbf{x})$, (*r*WDP) can be reduced to $\min_{\mathbf{x}\in X}f_1(\mathbf{x})$ which is actually a bi-level min–max problem.

Proposition 1. *(P2) is feasible and bounded from above.*

Proof. The feasibility of (P2) is trivial. To show (P2) is bounded from above, it is sufficient to show that the following terms in the objective function of (P2) are all bounded from above: $\sum_{i\in I}\bar{d}_i\lambda_{1,i},\sum_{j\in J}\sum_{k\in K_j}LS_{jk}x_{jk}\lambda_{2,jk}$ and $-\sum_{j\in J}\sum_{k\in K_j}US_{jk}x_{jk}\lambda_{3,jk}$. $\sum_{i\in I}\bar{d}_i\lambda_{1,i}$ is bounded from above since $\bar{d}_i$ is bounded and positive and $\lambda_{1,i}\leqslant e_i$ for all $i\in I$. $\sum_{j\in J}\sum_{k\in K_j}LS_{jk}x_{jk}\lambda_{2,jk}$ is bounded because $\lambda_{2,jk}$ is bounded for all $j\in J,k\in K_j$. $-\sum_{j\in J}\sum_{k\in K_j}US_{jk}x_{jk}\lambda_{3,jk}$ is bounded from above since $\lambda_{3,jk}$ is bounded from below for all $j\in J,k\in K_j$. □

Given Proposition 1, if in addition $X\neq\emptyset$, there exists an optimal solution to (*r*WDP).

Proposition 2. *There always exists an optimal solution to (P2) which is obtained at an extreme point of the feasible region of (P2).*

Proof. Due to the term $\bar{d}_i\lambda_{1,i}$ in Eq. (11), (P2) is bilinear. We denote the feasible region for all $\lambda_{1,i},\lambda_{2,jk},\lambda_{3,jk}$ by Λ, that is, $\Lambda\equiv\{(\lambda_{1,i},\lambda_{2,jk},\lambda_{3,jk},i\in I,j\in J,k\in K_j)|\ (12)\text{–}(15)\}$. It is apparent that (P2) is always feasible. $\bar{\mathbf{d}}$ and $(\boldsymbol{\lambda_1},\boldsymbol{\lambda_2},\boldsymbol{\lambda_3})$ are separately constrained. That is, the value of $\bar{\mathbf{d}}$ will not affect the feasible region of $(\boldsymbol{\lambda_1},\boldsymbol{\lambda_2},\boldsymbol{\lambda_3})$ and vice versa. Let's denote by $\bar{\mathbf{d}}^*$ and $(\boldsymbol{\lambda_1^*},\boldsymbol{\lambda_2^*},\boldsymbol{\lambda_3^*})$ an optimal solution to (P2) and let

$$f_2(\bar{\mathbf{d}},\boldsymbol{\lambda_1},\boldsymbol{\lambda_2},\boldsymbol{\lambda_3})=\sum_{i\in I}\bar{d}_i\lambda_{1,i}+\sum_{j\in J}\sum_{k\in K_j}(LS_{jk}x_{jk}\lambda_{2,jk}-US_{jk}x_{jk}\lambda_{3,jk}).$$

We have $f_2(\bar{\mathbf{d}}^*,\boldsymbol{\lambda_1^*},\boldsymbol{\lambda_2^*},\boldsymbol{\lambda_3^*})\geqslant f_2(\bar{\mathbf{d}},\boldsymbol{\lambda_1},\boldsymbol{\lambda_2},\boldsymbol{\lambda_3})$ for all $\bar{\mathbf{d}}\in\bar{\Delta}$ and all $(\boldsymbol{\lambda_1},\boldsymbol{\lambda_2},\boldsymbol{\lambda_3})\in\Lambda$. Note that the existence of such an optimal solution to (P2) is guaranteed given Proposition 1. If we fix $(\boldsymbol{\lambda_1},\boldsymbol{\lambda_2},\boldsymbol{\lambda_3})$ as $(\boldsymbol{\lambda_1^*},\boldsymbol{\lambda_2^*},\boldsymbol{\lambda_3^*})$ and solve (P2) for $\bar{\mathbf{d}}$, the problem reduces to a linear program and there must exist an optimal solution $\bar{\mathbf{d}}^{**}$ which is like $\bar{\mathbf{d}}^*$ also a basic feasible solution of $\bar{\Delta}$. As a result, we have $f_2(\bar{\mathbf{d}}^{**},\boldsymbol{\lambda_1^{**}},\boldsymbol{\lambda_2^{**}},\boldsymbol{\lambda_3^{**}})=f_2(\bar{\mathbf{d}}^*,\boldsymbol{\lambda_1^*},\boldsymbol{\lambda_2^*},\boldsymbol{\lambda_3^*})$. We then fix $\bar{\mathbf{d}}$ as $\bar{\mathbf{d}}^{**}$ and solve (P2) for $(\boldsymbol{\lambda_1},\boldsymbol{\lambda_2},\boldsymbol{\lambda_3})$ which is again a linear program and has an optimal solution $(\boldsymbol{\lambda_1^{**}},\boldsymbol{\lambda_2^{**}},\boldsymbol{\lambda_3^{**}})$ that is a basic feasible solution of Λ. So, we have

$$f_2(\bar{\mathbf{d}}^{**},\boldsymbol{\lambda_1^{**}},\boldsymbol{\lambda_2^{**}},\boldsymbol{\lambda_3^{**}})=f_2(\bar{\mathbf{d}}^{**},\boldsymbol{\lambda_1^*},\boldsymbol{\lambda_2^*},\boldsymbol{\lambda_3^*})=f_2(\bar{\mathbf{d}}^*,\boldsymbol{\lambda_1^*},\boldsymbol{\lambda_2^*},\boldsymbol{\lambda_3^*})\geqslant f_2(\bar{\mathbf{d}},\boldsymbol{\lambda_1},\boldsymbol{\lambda_2},\boldsymbol{\lambda_3})$$

for all $\bar{\mathbf{d}}\in\bar{\Delta}$ and all $(\boldsymbol{\lambda_1},\boldsymbol{\lambda_2},\boldsymbol{\lambda_3})\in\Lambda$. Consequently, $(\bar{\mathbf{d}}^{**},\boldsymbol{\lambda_1^{**}},\boldsymbol{\lambda_2^{**}},\boldsymbol{\lambda_3^{**}})$ is an extreme point of the feasible region $(\bar{\Delta}\times\Lambda)$ of (P2) and also the optimal solution to (P2). □

Based on Proposition 2, we can derive two different equivalent reformulations of $\min_{\mathbf{x}\in X}f_1(\mathbf{x})$, (P3) and (P4), both of which are single-level optimization problems.

$$\text{(P3)} \quad \min \quad A \tag{19}$$

$$\text{s.t.} \quad A \geqslant \sum_{i\in I} \bar{d}_i^l \lambda_{1,i}^l + \sum_{j\in J}\sum_{k\in K_j} \left(LS_{jk} x_{jk} \lambda_{2,jk}^l - US_{jk} x_{jk} \lambda_{3,jk}^l \right), \quad l \in \Omega \tag{20}$$

constraints (5)–(8)

In P(3), Ω is the index set of all the extreme points of $\bar{\Delta} \times \Lambda$ and A is a dummy decision variable.

$$\text{(P4)} \quad \min \quad B \tag{21}$$

$$\text{s.t.} \quad B \geqslant \sum_{j\in J}\sum_{k\in K_j} (c_{jk} y_{jk}^l + p_{jk} z_{jk}^l) + \sum_{i\in I} e_i \varphi_i^l, \quad l \in \bar{\Omega} \tag{22}$$

$$\sum_{j\in J}\sum_{k\in K_j} a_{ijk} y_{jk}^l + \varphi_i^l = \bar{d}_i^l, \qquad i \in I, \quad l \in \bar{\Omega} \tag{23}$$

$$LS_{jk} x_{jk} \leqslant y_{jk}^l + z_{jk}^l, \qquad j \in J, \quad k \in K_j, \quad l \in \bar{\Omega} \tag{24}$$

$$US_{jk} x_{jk} \geqslant y_{jk}^l, \qquad j \in J, \quad k \in K_j, \quad l \in \bar{\Omega} \tag{25}$$

$$y_{jk}^l, \quad z_{jk}^l, \quad \varphi_i^l \geqslant 0 \quad i \in I, \quad j \in J, \quad k \in K_j, \quad l \in \bar{\Omega} \tag{26}$$

constraints (5)–(8)

In P(4), $\bar{\Omega}$ is the index set of all the extreme points of $\bar{\Delta}$, B is a dummy decision variable, $y_{jk}^l, z_{jk}^l, \varphi_i^l$ are new recourse variables corresponding to $\bar{d}_i^l$. If we denote by $w_{\bar{\Delta}}$ and w_Λ the number of extreme points of $\bar{\Delta}$ and Λ, respectively, then $\|\Omega\| = w_{\bar{\Delta}} \cdot w_\Lambda$ and $\|\bar{\Omega}\| = w_{\bar{\Delta}}$.

4.3. Correlated volume of shipments

In many real-world cases of TL service auctions, the lane demands are highly correlated. Assume that $\tilde{d}_i,\ i \in I$ are correlated and there are independent and identically distributed random variables $\tilde{v}_n,\ n \in N,\ \|N\| < \|I\|$ with mean μ_v and variance σ_v^2. Specifically, we have $\tilde{\mathbf{d}} = \Xi \tilde{\mathbf{v}} + \tilde{\boldsymbol{\varepsilon}}$ where Ξ is an $\|I\| \times \|N\|$ matrix that describes the linear relationship between $\tilde{\mathbf{d}}$ and $\tilde{\mathbf{v}}$. To construct Ξ, one should first identify what $\tilde{\mathbf{v}}$ represent in reality (e.g., highway fee, macroeconomic conditions at the origin and destination of a lane). Then after collecting enough historical data of $\tilde{\mathbf{d}}$ and $\tilde{\mathbf{v}}$, we can evaluate Ξ by solving a set of linear equations of $\tilde{\mathbf{d}}$ and $\tilde{\mathbf{v}}$. $\tilde{\boldsymbol{\varepsilon}} = (\tilde{\varepsilon}_1, \ldots, \tilde{\varepsilon}_{\|I\|})$ are independent and identically distributed with mean 0 and variance σ_ε^2 and denotes random errors. Per Bandi and Bertsimas (2012), we can assume the possible realizations denoted by $\bar{d}_i,\ i \in I$ fall in the following uncertainty set:

$$\bar{\Delta} = \left\{ \bar{\mathbf{d}} | \bar{\mathbf{d}} = \Xi \bar{\mathbf{v}} + \bar{\boldsymbol{\varepsilon}}, \quad \left| \sum_{n\in N} \bar{v}_n - \|N\| \mu_v \right| \leqslant \Gamma \sigma_v \|N\|, \quad \left| \sum_{i\in I} \bar{\varepsilon}_i \right| \leqslant \Gamma \sigma_\varepsilon \|I\| \right\}$$

where Γ is a constant that usually takes the value of 2 or 3 to make a good fit empirically.

Following the steps in Section 4.2, we can reformulate (rWDP) as (P1) and derive the following max–min problem similar to (P2):

$$\text{(P2}') \quad \max \quad \left[\sum_{i\in I} \bar{d}_i \lambda_{1,i} + \sum_{j\in J}\sum_{k\in K_j} (LS_{jk} x_{jk} \lambda_{2,jk} - US_{jk} x_{jk} \lambda_{3,jk}) \right] \tag{11}$$

$$\text{s.t.} \quad \bar{d}_i = \sum_{n\in N} \Xi_{in} \bar{v}_n + \bar{\varepsilon}_i, \quad i \in I \tag{27}$$

$$\sum_{n\in N} \bar{v}_n \leqslant \mu_v \|N\| + \Gamma \sigma_v \|N\| \tag{28}$$

$$-\sum_{n\in N} \bar{v}_n \leqslant -\mu_v \|N\| + \Gamma \sigma_v \|N\| \tag{29}$$

$$\sum_{i\in I} \bar{\varepsilon}_i \leqslant \Gamma \sigma_\varepsilon \|I\| \tag{30}$$

$$-\sum_{i\in I} \bar{\varepsilon}_i \leqslant \Gamma \sigma_\varepsilon \|I\| \tag{31}$$

Constraints (12)–(15) and (18)

Let's represent (P2′) by $f_3(\mathbf{x})$ and (rWDP) can then be reduced to $\min_{\mathbf{x}\in X} f_3(\mathbf{x})$, a bi-level min–max problem. It is straightforward that Propositions (1) and (2) hold for (P2′) as well. Therefore, we can also derive (P3) and (P4) from (P2′). Since (P2′) is just a simple variation of (P2), both problems can be solved by the same method. For simplicity, in next section we demonstrate the solution methods under the assumption that shipping demands are independent and identically distributed.

5. Solution methods

As analyzed in the previous section, (rWDP) can be reduced to two equivalent single-level mathematical programs, (P3) and (P4), which however cannot be solved directly until all the extreme points of $\bar{\Delta} \times \Lambda$ and those of $\bar{\Delta}$ are enumerated, respectively. In most two-stage RO studies (Thiele et al., 2009; Jiang et al., 2012; Remli and Rekik, 2013; Bertsimas et al., 2013), the following Benders' decomposition type algorithm is applied to solve (P3).

Algorithm 1. Constraint Generation

Step 0. Set $\Omega = \emptyset$, $LB = -\infty$, $UB = +\infty$, $t = 1$. Go to Step 1.

Step 1. Solve (P3). Denote the optimal solution and the corresponding objective value as $\mathbf{x}^t$ and A^t, respectively. Set $LB = A^t$. Go to Step 2.

Step 2. Solve (P2) with first-stage decisions fixed as $\mathbf{x}^t$. Denote by $\left(\bar{d}_i^t, \lambda_{1,i}^t, \lambda_{2,jk}^t, \lambda_{3,jk}^t\right)$ the optimal solution. Set

$$UB = \min\left\{UB, \sum_{i\in I} \bar{d}_i^t \lambda_{1,i}^t + \sum_{j\in J}\sum_{k\in K_j}\left(LS_{jk}x_{jk}^t\lambda_{2,jk}^t - US_{jk}x_{jk}^t\lambda_{3,jk}^t\right)\right\}.$$

If $LB = UB$, stop the algorithm and $\mathbf{x}^t$ is the optimal first-stage solution. Otherwise, add t to the index set Ω, let $t = t + 1$ and go back to Step 1.

Note that in Step 2 of Algorithm 1, we need to solve the bilinear program (P2) which in general can only be solved to local optimum by classical gradient-based methods. However, a local solution to (P2) will under-estimate UB and may result in an early termination of Algorithm 1 with a less robust solution. Therefore, in order to guarantee the algorithm stops when the true solution to (P3) is found, we have to globally solve (P2) in every iteration of Algorithm 1.

Assuming a budget uncertainty set[7] and utilizing its special structure, the bilinear subproblem of a two-stage robust model can be reformulated as a mixed-integer program (MIP) which at a moderate scale can be globally solved by commercial MIP solvers such as CPLEX (Thiele et al., 2009; Gabrel et al., 2011; Remli and Rekik, 2013). However, this approach can not apply to (P2) directly.

Note that $\bar{\Delta}$ has $2\|I\|$ extreme points which can be explicitly expressed as:

$$\left.\begin{aligned} &\text{for } m = 1,\ldots,\|I\|, \quad \bar{d}_i^m = \begin{cases} \|I\|\mu + \Gamma\sigma\sqrt{\|I\|}, & i = m \\ 0, & i \neq m \end{cases} \\ &\text{for } m = \|I\|+1,\ldots,2\|I\|, \quad \bar{d}_i^m = \begin{cases} \max\left(\|I\|\mu - \Gamma\sigma\sqrt{\|I\|}, 0\right), & i = m - \|I\| \\ 0, & i \neq m - \|I\| \end{cases} \end{aligned}\right\} \tag{32}$$

Inspired by Proposition 2 and Eq. (32), we can reformulate (P2) as

$$\text{(P5)} \quad \max_{m=1,\ldots,2\|I\|} \left[\sum_{i\in I} \bar{d}_i^m \lambda_{1,i} + \sum_{j\in J}\sum_{k\in K_j}\left(LS_{jk}x_{jk}\lambda_{2,jk} - US_{jk}x_{jk}\lambda_{3,jk}\right)\right] \tag{33}$$
$$\text{s.t.} \quad \text{constraints}(12)-(15)$$

(P5) can be solved by the following algorithm.

Algorithm 2. Extreme Point Enumeration

Step 0. Set $z_0 = -\infty$, count = 0, $\bar{d}_i^* = 0, \lambda_{1,i}^* = 0, i \in I, \lambda_{2,jk}^*, \lambda_{3,jk}^* = 0, j \in J, k \in K_j$. Go to Step 1.

Step 1. For $m = 1,\ldots,2\|I\|$, evaluate $\bar{d}_i^m$ using Eq. (32), then solve

$$\text{(P6)} \quad \max \left[\sum_{i\in I} \bar{d}_i^m \lambda_{1,i} + \sum_{j\in J}\sum_{k\in K_j}\left(LS_{jk}x_{jk}\lambda_{2,jk} - US_{jk}x_{jk}\lambda_{3,jk}\right)\right] \tag{34}$$
$$\text{s.t.} \quad \text{constraints } (12)-(15)$$

[7] Budget uncertainty set is a special case of a polyhedral uncertainty set. We refer the readers to Bertsimas and Sim (2004) for more details.

Denote the objective value by z and the solution by $\left(\lambda^m_{1,i}, \lambda^m_{2,jk}, \lambda^m_{3,jk}\right)$. If $z \geqslant z_0$, let $\left(\bar{d}^*_i, \lambda^*_{1,i}, \lambda^*_{2,jk}, \lambda^*_{3,jk}\right) = \left(\bar{d}^m_i, \lambda^m_{1,i}, \lambda^m_{2,jk}, \lambda^m_{3,jk}\right)$.

Step 2. Output $\left(\bar{d}^*_i, \lambda^*_{1,i}, \lambda^*_{2,jk}, \lambda^*_{3,jk}\right)$ as the optimal solution to (P2).

By Algorithm 2, instead of solving the bilinear program (P2) at Step 2 of Algorithm 1, we actually solve $2\|I\|$ linear programs (P6), which guarantees obtaining a global optimal solution in polynomial time. Given that Λ is also a polyhedron and thus contains a finite number of extreme points, we have the following proposition.

Proposition 3. *If (P3) can be globally solved at Step 1 in each iteration of Algorithm 1, (rWDP) can be globally solved by Algorithm 1 embedded with Algorithm 2 in a finite number of iterations.*

Due to (32), we can also reformulate (P4) as:

(P7) $\min B$

$$\text{s.t.} \quad B \geqslant \sum_{j\in J}\sum_{k\in K_j}\left(c_{jk}y^m_{jk} + p_{jk}z^m_{jk}\right) + \sum_{i\in I} e_i\varphi^m_i, \quad m = 1,\ldots,2\|I\| \tag{35}$$

$$\sum_{j\in J}\sum_{k\in K_j} a_{ijk}y^m_{jk} + \varphi^m_i = \bar{d}^m_i, \quad i \in I, m = 1,\ldots,2\|I\| \tag{36}$$

$$LS_{jk}x_{jk} \leqslant y^m_{jk} + z^m_{jk}, \quad j \in J, k \in K_j, m = 1,\ldots,2\|I\| \tag{37}$$

$$US_{jk}x_{jk} \geqslant y^m_{jk}, \quad j \in J, k \in K_j, m = 1,\ldots,2\|I\| \tag{38}$$

$$y^m_{jk}, z^m_{jk}, \varphi^m_i \geqslant 0 \quad i \in I, j \in J, k \in K_j, m = 1,\ldots,2\|I\| \tag{39}$$

constraints (5)–(8)

where the value of $\bar{d}^m_i$ is determined by (32). Note that (P7) is a single-level MIP which at a moderate scale can be solved by commercial MIP solvers.

Now, we have two methods to solve (*r*WDP): (a) applying a Benders' decomposition type constraint generation algorithm embedded with Extreme Point Enumeration; and (b) solving an equivalent MIP reformulation of (*r*WDP). For simplicity, in the rest of this paper, we call method (a) constraint generation (CG) and method (b) solving reformulation (SR). Both methods involve solving MIPs and can converge to a global optimum in a finite number of iterations if those MIPs can be globally solved. We will compare the performance of these two methods in next section.

6. Numerical Experiments

In this section, we conduct numerical experiments to demonstrate the tractability of our proposed robust optimization method and test the quality of the robust solution. Specifically, in Section 6.1, we compare the efficiency of the two solution methods, CG and SR, developed in the previous section. In Section 6.2, we compare numerical tractability of the robust WDP and (*d*WDP). In addition, the quality of the robust solution is demonstrated by simulation.

All the tests are coded in GAMS 23.6.5 and solved using CPLEX 12.4 solver. We run the tests on the Penn State Lion-XC system, which is a computational cluster run by the Penn State High Performance Computing Group, with the following attributes: AMD Opteron250 2.4 GHz and 8 GB RAM.

6.1. Comparison of CG and SR

Both CG and SR can converge to a global solution if the MIPs involved are globally solved. So we are interested only in comparing the numerical tractability of these two methods. Numerical tests are conducted on WDPs at five different scales in [number of lanes, number of bidders, number of packages submitted per bidder]: [10, 5, 5], [30, 6, 10], [60, 10, 10], [120, 15, 10] and [180, 20, 10]. In each scenario, the bids submitted by each bidder which are reflected by the value of a_{ijk} are randomly generated. The unit shipping cost quoted on each package c_{jk} is evaluated by multiplying the number of lanes included in the package by a random number in [0, 1/(total number of lanes in the auction)] so that the average unit shipping cost on every single lane has the same distribution. Penalty costs p_{jk} double the corresponding unit shipping costs. The unit extra service costs e_i are also randomly generated so that on a lane the average extra service cost is 10/(total number of lanes in the auction) more than the average unit shipping cost. We assume that the volume of shipments on each lane $\tilde{d}_i$ is i.i.d. with mean 100 and variance 100. The lower bound and upper bound on acceptable shipment volume proposed by each carrier on each bid, LS_{jk} and US_{jk}, are uniformly distributed in [25, 75] and [125, 175], respectively. The minimum and maximum number of winners of the auction are 2/5 and 4/5 of the total number of bidders, respectively.[8] We set $\Gamma = 3$.

[8] If the number is not an integer, we will round it to the smallest following integer.

The result is summarized in Table 2.

In Table 2, "gap" denotes the relative gap between the best possible objective value and the objective value obtained when the solver terminates. In column "time", the number refers to the computation time which is counted in seconds. "NA" indicates that the solver terminates because it runs out of memory. Given Table 2, it is apparent that SR is much more efficient than CG in solving the two-stage robust winner determination model. For instance, CG can not solve a problem with more than 60 lanes while SR only uses 12.38 s to solve the 180-lane problem to its global optimum. It takes CG about 1.5 h to solve the 30-lane problem while SR solves the same problem in 0.15 s. In the following tests, we adopt SR as the solution approach to the proposed two-stage robust winner determination model.

6.2. Comparison with Deterministic Solution

Given mean and variance of the uncertain shipment volume, an alternative solution to the WDP is to solve a deterministic model using the mean as an estimate for shipment volume. We compare our proposed two-stage robust optimization approach with this deterministic approach in this section. For simplicity, in the rest of this section, we denote the former approach by "RO-SR" and the later one by "DS".

6.2.1. Tractability

We first compare the tractability of RO-SR and DS. Again, we test on the same problems shown in Section 6.1. The result is summarized in Table 3.

From Table 3, we can see that both RO-SR and DS can solve the smallest 10-lane problem in around 0.18 s. But as the problem scale increase, the computation time required by DS increases dramatically and more significantly than that required by RO-SR. To solve the 120-lane problem, the time required by DS and RO-SR are 5755.01 s and 3.80 s, respectively. DS can not solve the 180-lane problem due to lack of memory while RO-SR can solve the problem at the same scale in 12.38 s. To sum up, RO-SR is a much more efficient method than DS. This looks surprising at first because although RO-SR and DS both solve single-level MIPs, the one solved by RO-SR, which is (P7), has more demand consumption constraints and second-stage decision variables than that solved by DS, which is (dWDP). By examining the formulation of (P7), we can find the following two causes of such a surprising result. First, most demand consumption constraints in (P7) are redundant due to zero right-hand-sides and many second-stage decision variables are forced to be 0. Second, in those demand consumption constraints that are not redundant, many second-stage decision variables are also forced to be 0. To elaborate more about the latter cause, let's consider a simple example. Assume that $\|I\|\mu - \Gamma\sigma\sqrt{\|I\|} \leqslant 0$, $a_{1jk} = 1$ only when $j = 1$, $k = 1, 2$ and $a_{2jk} = 1$ only when $j = 1$, $k = 2$. In deterministic model, the demand consumption constraints when i = 1, 2 are:

$$\begin{cases} y_{11} + y_{12} + \varphi_1 = d_1 \\ y_{12} + \varphi_2 = d_2 \end{cases}$$

In (P7), the demand consumption constraints when $i = 1, 2$ are:

$$\begin{cases} y_{11}^1 + y_{12}^1 + \varphi_1^1 = \|I\|\mu + \Gamma\sigma\sqrt{\|I\|} \\ y_{11}^m, y_{12}^m, \varphi_1^m = 0, \quad \forall m \neq 1 \\ y_{12}^2 + \varphi_2^2 = \|I\|\mu + \Gamma\sigma\sqrt{\|I\|} \\ y_{12}^m, \varphi_2^m = 0, \quad \forall m \neq 2 \end{cases}$$

which obviously can be reduced to

$$\begin{cases} y_{11}^1 + \varphi_1^1 = \|I\|\mu + \Gamma\sigma\sqrt{\|I\|} \\ \varphi_2^2 = \|I\|\mu + \Gamma\sigma\sqrt{\|I\|} \end{cases}$$

In (P7), the demand consumption constraints actually contain less decision variable than in deterministic model.

6.2.2. Solution quality

Although robust optimization provides solutions that are generally able to handle uncertainty, it is also recognized as a conservative method because it focuses on optimizing the objective value only in the worst-case scenario. Moreover, in our two-stage robust winner determination model, the uncertainty set constructed using mean and variance of the shipment

Table 2
Comparison of CG and SR in tractability.

	10		30		60		120		180	
	Gap (%)	Time	Gap (%)	Time	Gap (%)	Time	Gap (%)	Time	Gap (%)	Time
CG	0.0	34.28	0.0	5218.80	3.0	NA	12.3	NA	9.2	NA
SR	0.0	0.18	0.0	0.15	0.0	0.66	0.0	3.80	0.0	12.38

Table 3
Comparison of DS and RO-SR in tractability.

	10		30		60		120		180	
	Gap (%)	Time	Gap (%)	Time	Gap (%)	Time	Gap (%)	Time	Gap (%)	Time
DS	0.0	0.19	0.0	1.06	0.0	16.82	0.0	5755.01	13.2	NA
RO-SR	0.0	0.18	0.0	0.15	0.0	0.66	0.0	3.80	0.0	12.38

Table 4
Comparison of DS and RO-SR in terms of solution quality.

Variance	DS	RO-SR	Improvement (%)
600	930.16	1210.36	−30.1
9000	1485.28	1239.95	16.5

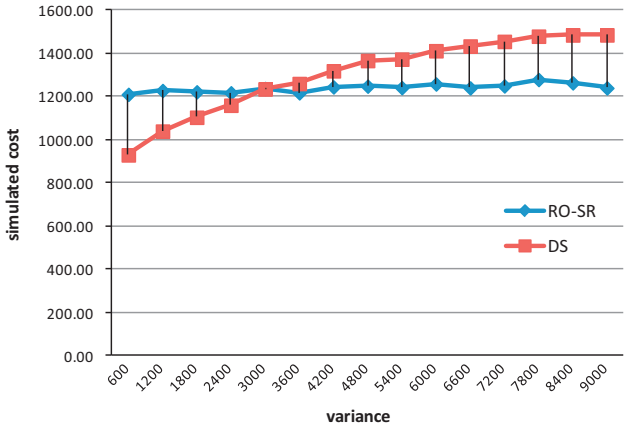


Fig. 2. Comparison of DS and RO-SR in terms of solution quality at different levels of variance.

volume is a general polyhedral set. Thus, intuitively, our model tends to provide conservative solutions. In this section, we test the quality of solutions derived by RO-SR using simulation and compare it with the solution derived by DS.

The experiments are designed as follows. Given the mean and variance of the volume of shipments on each lane, we solve the WDP by DS and RO-SR and we get two different solutions for the first-stage decision variables. Then we generate 100 random volumes of shipments on each lane and obtain 100 simulated objective values (procurement costs) by solving the second-stage of WDP. The average of 100 simulated objective values is used to compare the quality of solution derived by DS and RO-SR.

The experiments are conducted on the 60-lane problem which is introduced in Section 6.1. The mean is again assumed to be 100. We test the quality of two solutions at different levels of variance. We assume Gamma distribution for the simulation. The results when the variance is small (600) and large (9000) are summarized in Table 4.

According to Table 4, when the variance is small, DS is a better approach than RO-SR since the average simulated cost associated therewith is less. Indeed, RO-SR is conservative compared to DS when the variance is small. However, when the variance is large, RO-SR provides a better solution than DS. As shown by Fig. 2, as the variance of the uncertainty increases, the average simulated cost associated with DS solution increases significantly while that associated with RO-SR solution does not change significantly with the variance.

Then we consider the case when the lane demands are correlated and demonstrate how incorporating correlation information can reduce the procurement cost. The experiment is designed in a similar way. In this case, we denote our approach by "LC-SR". In addition to DS, we also compare LC-SR with the robust method using budget uncertainty set (Remli and Rekik, 2013) and denote the latter by "BU-SR".

To create a covariance matrix, we first randomly generate a symmetric matrix and then adjust its scale to make sure that the sum and sum of square of the elements in each row are fixed as 1 and 10, respectively. We assume that the elements of both $\tilde{\mathbf{v}}$ and $\tilde{\boldsymbol{\varepsilon}}$ follow normal distribution and set $\sigma_{\tilde{\varepsilon}}^2 = 10$. We adjust the mean and variance of each element in $\tilde{\mathbf{v}}$ to make sure that the mean of shipping demand on each lane is 100 and the variance increases step by step from 200 to 7000. For BU-SR, we assume that on each lane the shipping demand falls into [75, 125] and the sum of all demands on 60 lanes falls into [5000, 7000]. We use Gamma distribution for the simulation. The results are summarized in Table 5.

Table 5
Comparison of DS, LC-SR, and BU-SR in terms of solution quality.

Method	Variance = 200	Improvement (%)	Variance = 7000	Improvement (%)
DS	863.18	0.0	2582.85	0.0
LC-SR	1114.72	−29.14	1484.20	42.54
BU-SR	1317.25	−52.60	1678.95	35.00

From Table 5, we can see that when the variance is small, DS outperforms both LC-SR and BU-SR. However, it turns around when the variance is large. Specifically, LC-SR outperforms BU-SR in both cases. We also plot the average simulated cost at different levels of variance in Fig. 3. We can see that the cost associated with DS solution increases significantly while those associated with BU-SR and LC-SR solutions do not change much with the variance, as the variance of shipment volume increases. Fig. 3 also shows that the LC-SR always outperforms BU-SR under correlated lane demand uncertainty.

7. Further discussion

In Section 3, we assumed that if a carrier wins a package bid, the volume of shipments on each lane in the package will be the same. In this section, we relax this assumption and discuss its effects on the proposed robust winner determination model.

The main structure of the combinatorial auction for TL service procurement is the same. The only difference is that carriers will specify the bounds of acceptable volume of shipments and quote shipping rate on each single lane included in a package bid. Correspondingly, the volume of shipments assigned to a carrier on each lane could be different and if they are outside the specified acceptable interval a lane-specific penalty will be charged to the shipper.

To avoid confusion, the following new notations in Table 6 are used to model the relaxed deterministic WDP (*e*WDP).

The relaxed model can be formulated as:

$$(e\text{WDP}) \quad \min \sum_{i\in I}\left[\sum_{j\in J}\sum_{k\in K_j}(\bar{c}_{ijk}\bar{y}_{ijk}+\bar{p}_{ijk}\bar{z}_{ijk})+e_i\varphi_i\right] \tag{40}$$

$$\text{s.t.} \quad \sum_{j\in J}\sum_{k\in K_j}\bar{y}_{ijk}+\varphi_i=d_i, \quad i\in I \tag{41}$$

$$\overline{LS}_{ijk}x_{jk}\leqslant \bar{y}_{ijk}+\bar{z}_{ijk}, \qquad i\in I, \quad j\in J, \quad k\in K_j \tag{42}$$

$$\overline{US}_{ijk}x_{jk}\geqslant \bar{y}_{ijk}, \qquad i\in I, \quad j\in J, \quad k\in K_j \tag{43}$$

$$\bar{y}_{ijk},\bar{z}_{ijk},\varphi_i\geqslant 0 \qquad i\in I, \quad j\in J, \quad k\in K_j \tag{44}$$

constraints (5)−(8)

We can then formulate the corresponding two-stage robust WDP:

$$(r\text{WDP}') \quad \min_{\mathbf{x}\in X}\max_{\mathbf{d}\in\Delta}\min_{(\bar{\mathbf{y}},\bar{\mathbf{z}},\varphi)\in\bar{Y}(\mathbf{x},\mathbf{d})} f_4(\bar{\mathbf{y}},\bar{\mathbf{z}},\varphi) \tag{45}$$

where

$$f_4(\bar{\mathbf{y}},\bar{\mathbf{z}},\varphi)=\sum_{i\in I}\left[\sum_{j\in J}\sum_{k\in K_j}(\bar{c}_{ijk}\bar{y}_{ijk}+\bar{p}_{ijk}\bar{z}_{ijk})+e_i\varphi_i\right],$$

Table 6
New notations.

Symbol	Description
$\overline{LS}_{ijk}$	The minimum volume of shipments on lane i in package k that is required by carrier j if carrier j wins package k, if lane i is not included in carrier j's package $k, \overline{LS}_{ijk}=0$
$\overline{US}_{ijk}$	The maximum volume of shipments on all lanes in package k that is required by carrier j if carrier j wins package $k, \overline{US}_{ijk}\geqslant\overline{LS}_{ijk}$ and if lane i is not included in carrier j's package $k, \overline{US}_{ijk}=0$
p'_{ijk}	The penalty cost of being in short of 1 unit of shipment on lane i in winner j's package k, if lane i is not included in carrier j's package k, p'_{ijk} equals a large number
c'_{ijk}	The price of shipping 1 unit of freight on lane i quoted by carrier j on their package k, if lane i is not included in carrier j's package $k, \bar{c}_{ijk}$ equals a large number
$\bar{y}_{ijk}$	Volume of shipments assigned to carrier j on lane i in their package k
$\bar{z}_{ijk}$	The shortage in required shipment volume on lane i in carrier j's package k

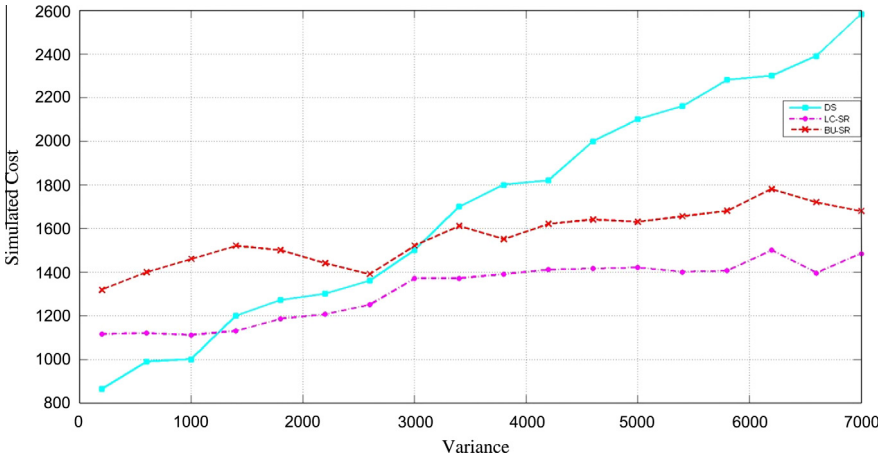


Fig. 3. Comparison of DS, LC-SR and BU-SR in terms of solution quality at different levels of variance.

$$X = \{(x_{jk}, j \in J, k \in K_j)t|\ (5)-(8)\},$$

$$\overline{Y}(\mathbf{x}, \bar{\mathbf{d}}) = \{(\bar{y}_{ijk}, \bar{z}_{ijk}, \varphi_i, i \in I, j \in J, k \in K_j)|\ (41)-(43),\ (45)\}.$$

Per strong duality theory, the inner minimization problem can be replaced by its dual:

$$(\mathrm{P1}') \quad \max \quad \left[\sum_{i\in I}(\bar{d}_i\bar{\lambda}_{1,i} + \sum_{j\in J}\sum_{k\in K_j}(\overline{LS}_{ijk}x_{jk}\bar{\lambda}_{2,ijk} - \overline{US}_{ijk}x_{jk}\bar{\lambda}_{3,ijk}))\right] \tag{46}$$

$$\text{s.t.} \quad \bar{\lambda}_{1,i} + \bar{\lambda}_{2,ijk} - \bar{\lambda}_{3,ijk} \leqslant \bar{c}_{ijk}, \qquad i \in I, \quad j \in J, \quad k \in K_j \tag{47}$$

$$\bar{\lambda}_{1,i} \leqslant e_i, \quad i \in I \tag{48}$$

$$\bar{\lambda}_{2,ijk} \leqslant \bar{p}_{ijk}, \qquad i \in I, \quad j \in J, \quad k \in K_j \tag{49}$$

$$\bar{\lambda}_{2,ijk}, \bar{\lambda}_{3,ijk} \geqslant 0, \qquad i \in I, \quad j \in J, \quad k \in K_j \tag{50}$$

where $\bar{\lambda}_{1,i}, \bar{\lambda}_{2,ijk}, \bar{\lambda}_{3,ijk}$ are the dual variables corresponding to constraints (41)–(43), respectively. Then the inner max–min problem can be reformulated as a maximization problem:

$$(\mathrm{P2}'') \quad \max \quad \left[\sum_{i\in I}(\bar{d}_i\bar{\lambda}_{1,i} + \sum_{j\in J}\sum_{k\in K_j}(\overline{LS}_{ijk}x_{jk}\bar{\lambda}_{2,ijk} - \overline{US}_{ijk}x_{jk}\bar{\lambda}_{3,ijk}))\right] \tag{46}$$

$$\text{s.t.} \quad \text{constraints } (16)-(18), (47)-(50)$$

We represent (P2″) by $f_5(\mathbf{x})$. (rWDP′) can be reduced to $\min_{\mathbf{x}\in X} f_5(\mathbf{x})$ which is a bi-level min–max problem.

Following the ideas of a series of derivations in Section 4.2, we can easily find that Propositions (1) and (2) also hold for (P2″). Thus we can derive (P3) and (P4) from (P2″) and solve (P2″) by the proposed approach. As a result, the proposed robust winner determination model applies to the relaxed problem as well.

8. Conclusion

In this paper we proposed a new tractable two-stage robust winner determination model for truckload service procurement under shipment volume uncertainty. Specifically, we refine the deterministic model by penalizing shortages and relaxing the constraint that prohibits multiple winners of each lane. The modified model is more general and has a larger feasible region under shipment volume uncertainty. Moreover, we argue that in practice exact distributional information of uncertainty may not be available and propose a robust winner determination model utilizing only partial distributional information of uncertain shipment volume. In particular, we construct a general polyhedral uncertainty set utilizing mean and variance of the uncertain shipment volume and formulate a two-stage robust winner determination model. Although robust optimization is recognized as a conservative method which provides solution only optimal in the worst-case scenario, we use simulation-based numerical tests to show that the solution to our robust WDP attains low procurement cost in general scenarios too, especially when the variance of the uncertainty is large. In particular, the simulated procurement cost associated with the robust solution does not change significantly with the variance of the shipment volume uncertainty, while

the cost associated with the deterministic solution increases dramatically as the uncertain shipment volume becomes more variant. More importantly, our proposed approach can handle correlated lane demands and numerical results demonstrate the value of it.

WDP is an NP-hard problem and is challenging to solve even without uncertainty. In this paper, we develop a solution method which is more efficient than the widely adopted Benders' type constraint generation algorithm. Moreover, we demonstrate that our robust WDP has better numerical tractability than a deterministic WDP, especially when the problem scale is large.

An interesting topic for future research is to balance tractability and solution quality. As we show in this study, although the proposed two-stage robust model is numerically tractable, it is conservative when the variance of uncertainty is small. Assuming more knowledge of the uncertainty will reduce the conservativeness of the robust model. However, its impact on problem tractability remains to be investigated. Lastly, incorporating game-theoretic behaviors of shippers and carriers in truckload service procurement under uncertainty is another interesting research direction. In particular, new bidding structures for the auction or new mechanisms which can further reduce shippers' truckload transportation procurement cost is worth investigating.

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Appendix A. The construction of uncertainty set when lane demands are independent but not identically distributed

Assume that $\tilde{d}_i,\ i \in I$ are independent but not identically distributed random variables with mean μ_i and variance σ_i^2. Define $s_I^2 = \sum_{i\in I}\sigma_i^2$. Assume that for some positive ζ, the follow equation holds:

$$\lim_{\|I\|\to+\infty} \frac{1}{s_I^{2+\zeta}} \sum_{i\in I} E\left[\left|\tilde{d}_i - \mu_i\right|^{2+\zeta}\right] = 0.$$

Then as $\|I\| \to +\infty, \sum_{i\in I}(\tilde{d}_i - \mu_i)/s_I$ follows a standard normal distribution (see Theorem 27.3 in Billingsley (1986)). That is,

$$\lim_{\|I\|\to+\infty} Pr\left(\frac{\sum_{i\in I}(\tilde{d}_i - \mu_i)}{\sqrt{\sum_{i\in I}\sigma_i^2}} \leqslant t\right) = Pr(Z \leqslant t)$$

where $Z \sim N(0,1)$. Then we can assume the possible realizations denoted by $\bar{d}_i,\ i \in I$ fall in the following uncertainty set:

$$\Delta = \left\{(\bar{d}_1, \ldots, \bar{d}_{\|I\|}) \middle| \left|\sum_{i\in I}(\bar{d}_i - \mu_i)\right| \leqslant \Gamma\sqrt{\sum_{i\in I}\sigma_i^2}\right\}$$

where Γ is a constant that usually takes the value of 2 or 3 to make a good fit empirically.

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